20
42
62%

KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCES

COURSE: SYSTEM VULNERABILITY ASSESSMENT **COURSE CODE: CYS 330**
DATE: 21ST MARCH 2025 **TASK: TEST ONE**

Instructions

1. Write your **Student Number** on the **Answer Sheet (Booklet)** provided
2. Write your program name on the **Answer Sheet (Booklet)** provided
3. There are **four (4)** questions in this paper. Marks are allocated at the end of each question. Each question carries **25 Marks.**
4. You are allowed to have an allowance of **5 minutes** to go through your paper before the beginning of the Test. Note that a good hand-writing is encouraged.
5. Total marks for this paper: 100%
6. Duration: **2 Hours** 30 Min

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Question One [25 Marks]

1. A system can be considered to be insecure and unsecure and therefore, susceptible to vulnerabilities.
   - **(a)** Define the following system vulnerability terms:
     - **(i)** Insecure system [2] ✓
     - **(ii)** Unsecure system [2] ✓
     - **(iii)** Mitigation [2] ✓
     - **(iv)** Remediation [2] ✓
     - **(v)** Attack surface [2] ✓

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   - **(b)** Give **two (2)** examples of the following system vulnerability terms:
     - **(i)** Insecure systems [2] ✓
     - **(ii)** Unsecure systems [2] ✓
   - **(c)** Provide mitigation solutions to answers given in **(b);**
     - **(i)** Insecure systems [2] ✓
     - **(ii)** Unsecure systems [2] ✓

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   - **(d)** Provide remediation solutions to answers given in **(b)**
     - **(i)** Insecure systems [2] ✓
     - **(ii)** Unsecure systems [2] ✓
2. Mention and explain **three (3)** major differences between mitigation and remediation [3] ✓

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Question Two [25 Marks]

1. List and explain **five (5)** reasons why security might fail in an organisation [5] ✓

   5
2. in a table format, differentiate between Vulnerability Assessment (VA) and Vulnerability Management (VM) [12] ✓

   12
3. Discuss in detail, the following vulnerabilities
   - **(a)** Buffer overflow/overrun [2] ✓
   - **(b)** Unpatched software [2] ✓
   - **(c)** Cross-Site Request Forgery (CSRF/XSRF) [2]
   - **(d)** Eavesdropping (sniffing/snooping) [2] ✓

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CamScanner

$\frac{1}{2} = 0.5$

$0 \times 2 = 0 - 1$

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Question Three [25 Marks]

1. Most devastating vulnerabilities occur at the level of system software. Describe **two (2)** probable vulnerabilities associated with the following critical software and also, give probable solutions to vulnerabilities accordingly;
   - **(a)** Operating system [4]
   - **(b)** Firmware [4]
   - **(c)** Device drivers [4]
   - **(d)** Middleware [4]
   - **(e)** Utility [4]

   rootkit outdated → Missing

2. A named bank in Zambia has halted online payments by a named University because they have discovered vulnerabilities in the API. Luckily, you have been engaged to solve this problem considering your expertise in system vulnerability assessment and management. Give and explain in detail, five (5) ways how you would assess for vulnerabilities in the API. [5]

   missing outdated

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Question Four [25 Marks]

1. History has shown that attackers could even attack the Integrated circuits that make up the computer architecture. One of the vulnerabilities to talk about here is the hardware trojan.
   - **(a)** Define a hardware trojan [2]
   - **(b)** List and explain two characteristics of hardware trojans [2]
   - **(c)** Explain in detail, giving three (3) possible impact of HT [3]
   - **(d)** Von Neumann describe the CPU as one of the most important part of the computer. It is vulnerable to hardware trojans because it is an IC meaning, it uses logic gates and circuits. This statement makes the CPU vulnerable to attacks.
     Represent the following logic gates and their truth tables;
     - **(i)** OR gate [3]
     - **(ii)** AND gate [3]
     - **(iii)** NOT gate [2]
     - **(iv)** Discuss theft of service attack vulnerability in the OS/CPU. [2]
2. There is a common say in cyber security which says 'you cannot protect what you do not know'. Some services provided by the cloud include such services as PaaS, SaaS, CaaS, IaaS, and FaaS. Therefore, it is imperative that you have knowledge on cloud computing. Write short notes on Cloud computing in terms of;
   - **(i)** Scheduler vulnerabilities in Cloud computing [2]
   - **(ii)** Example of a vulnerability in a Hypervisor and it's impact in the virtual Machine environment (cloud) [2]
   - **(iii)** Timing attacks in the scheduling algorithm [2]
   - **(iv)** Explain briefly, how you could secure the Cloud as an expert [2]

Q1 = 8+8+3 = 19

Q2 = 5+12+6 = 23

Q3 = 4+5 = 9

Q4 = 7+4 = 11

Total = 19 + 23 = 42

= 42 + 29 = 51

= 51 + 11 = 62

CamScanner